Morehouse HJ Political Science Lab

Final Report: Open Critical Mapping Workshop

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Teaching Assistants:

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- Andrew Ratcliff, Senior, Political Science Major, Morehouse College
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- Mali Glemaud Thesee, Senior, Computer Science Major, Morehouse College

Project GitHub: https://github.com/jjang52/open-mapping



I. Purpose of the Project

This project was conceptualized as an opportunity to teach faculty and students how to geographically map open-source census data variables using R. The variables can be mapped on states, cities, tracts, and even other countries. The project also presented the opportunity to overlay multiple variables over the same map at once to find geographic relationships via the spatial data. Our collective efforts allowed faculty and students in the Atlanta University Center (AUC) Consortium to get hands-on training in R and incorporate the techniques into their individual research projects. The sessions were broken up into two segments; session one, exclusively for faculty of the college, and session two, for student participants. The faculty session was led by the HJ Political Science Lab Principal Investigator (PI), Dr. Hye Ryeon Jang, Assistant Professor of Political Science at Morehouse College. The lab's teaching assistants were prepared to assist if faculty members encountered roadblocks. The session was led using an R markdown (RMD) file which used median household income and Black population census data to illustrate gentrification in Fulton County in Atlanta, GA. This served as an example for the faculty to help visualize how their research interests may intersect with the techniques taught in the session.

The second session was led by the teaching assistants of the HJ Lab, and featured 13 students from Morehouse College, Spelman College, and Clark Atlanta University, as participants. The session was centered around the fundamentals of R, as well as the interpretation of the exemplary RMD file. By the end, students were assigned new variables, and tasked with geographically inserting them onto Fulton County maps.

The rest of this report will provide more specifics about the respective sessions, including faculty testimonies and student survey data detailing their growth from the lab session. Lastly, this report will explain how students and faculty alike intend to use their newfound knowledge for their own research endeavors.



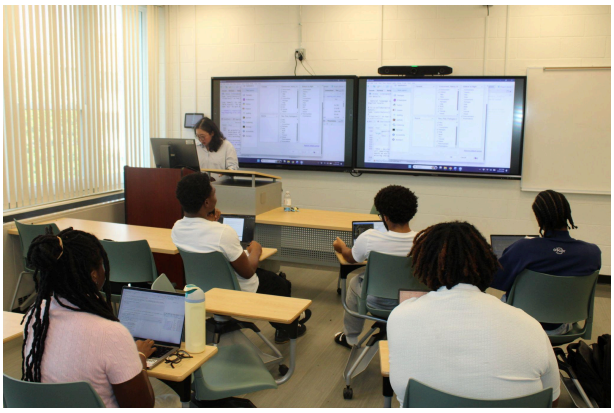


Figure 1 and 2: Dr. Hye Ryeon Jang (HJ Lab Principal Investigator) training the teaching assistants in using open-source census data and spatial data to make critical map

II. Procedure

Faculty Workshop

The Open Critical Mapping faculty sessions took place on September 28 and 30, 2026, at Morehouse College. The overall sessions consisted of 10 faculty participants from political science, sociology, world languages and culture, business administration, psychology (*Figure 3*).

The most rudimentary part of the Open Critical Mapping effort was installation of R, and Rstudio, as well as the retrieval of Census API keys, from census.gov, which all participants completed prior. This step is vital in ensuring users of R are able to manipulate, assess, and analyze data published by the Census bureau. Next, participants were required to install, and load the *tidycensus*, *tidyverse*, *sf*, *virdis*, and *mapview* packages which are crucial in efficiently completing the computational geographical techniques expelled in the workshop. With the previously attained API keys, participants were instructed to insert them into their personal RMD file previously provided by the HJ Lab, granting them access to manipulate the census data. Using data from 2023 for completeness, participants were tasked with parsing through the ACS data, recording demographic variables potentially relevant to their research projects.

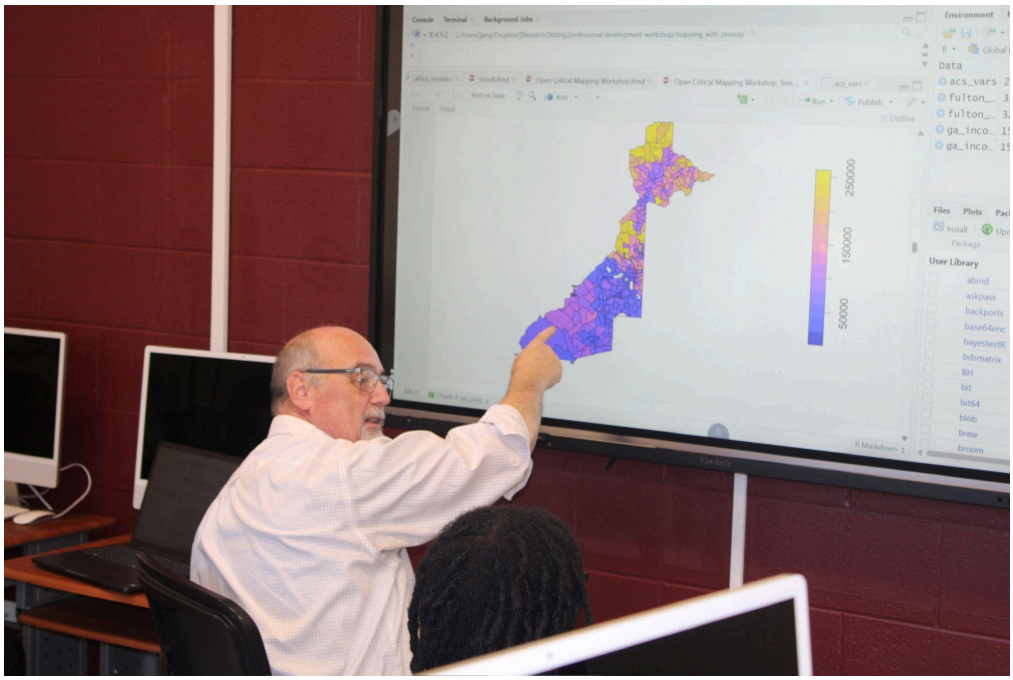


Figure 3: Dr. Keith Hollingsworth interpreting the map of Fulton county median household income.

Following participants gathering relevant demographic variables, we began the process of generating maps which displayed the variables participants selected. In our workshop, the exemplar RMD file generated maps displaying income in the state of Georgia, and Fulton County, however participants were encouraged to generate original maps of their hometowns, or areas relevant to their research projects.

Next, using the *ggplot* and *mapview* functions, we encouraged participants to generate more detailed, and interactive maps, by adding tables on the X and Y axis, a map legend, altering color opacity and adding a title. Using *mapview*, participants were instructed on how to produce an interactive map, allowing them to pinpoint data from specific census tracts within the geographic boundary. Finally, users were introduced to computational geography: layering two variables onto the same map. In the exemplary RMD, we mapped the Black population and income of Fulton County, Georgia onto the same map. Furthermore, participants were instructed on the process of mapping gentrification scores within a geographical boundary. The process of extracting the gentrification score of an area is crucial in not only understanding where gentrification takes place, but developing further research hypotheses and theoretical approaches behind the causes.

The pedagogical approach to mapping introduced by the HJ Lab provided an efficient alternative at generating digital cartography which most participants will utilize well beyond the workshops. As reported by Dr. Hollingsworth (Business Administration), this alternative approach to mapping will be vital in increasing the efficiency for delineating the transformation of Black businesses in Atlanta, Georgia. After completing the ORCA workshop, testimony reported by Dr. Chang (Psychology) displays a transformative experience similar to Dr. Hollinsworth. Dr. Chang emphasized the importance of further intertwining computational geography within the investigation of the psychological motivations behind community engagement with public goods. On a larger scale, Dr. Chang mentioned her consideration of moving away from instruction using SPSS and transitioning to RStudio, accrediting her attendance of the workshop in influencing her decision.

Student Workshop

The student session was conducted by the lab's teaching assistants on October 2, 2026. This was an opportunity for the teaching assistants to develop mastery over R as they were tasked with instructing the attendees the fundamental principles of the coding software (*Figure 4*). The session was attended by 14 students from Morehouse College, Spelman College, and Clark Atlanta University.

The attendees were first guided to install R and RStudio if they had not already. Then, they retrieved their API keys and were explained the basics of R. The same RMD file that was used during the faculty workshop was used as the demonstrative guide for the student workshop. Five variables were explained to measure gentrification in different ways: median household income, Black population, population of high school diploma holders, unemployed population, population of impoverished white people, and population of impoverished Black people.

As the workshop proceeded, individual lines of the code were explained to the students, so they understood what they were typing and not simply going through the motions haphazardly. The first major code chunk allowed for the students to call the median household income variable and make an object which links the variable to Georgia's geography. They did the same with Fulton county. Then those objects were run using spatial data which added the longitudinal and latitudinal measurements within the state. Finally, the students ran the code and created maps using the objects they just made.

They were then provided with some tips on how to clean up their maps visually by adding labels, titles, grid lines with the geographic measurements labeled on the axes, as well as tips on how to diversify the color gradients of their maps. The last step of the workshop demonstration showed students how to overlay a second variable onto the map. For the session's example, the TAs map illustrated the Black population in Fulton County and median household income. This was divided up between tracts, and one red dot on the map represented 1,000 Black people in that tract.



Figure 4: Zion Spruill (TA) circulating the room during the student workshop, providing support when necessary, ensuring students did not fall behind.

Throughout this demonstration, one or two TAs were at the front of the room (*Figure 5*), navigating the RMD file which was projected onto the TV while the remaining five TAs circulated the room, ensuring that everyone was on the same pace.

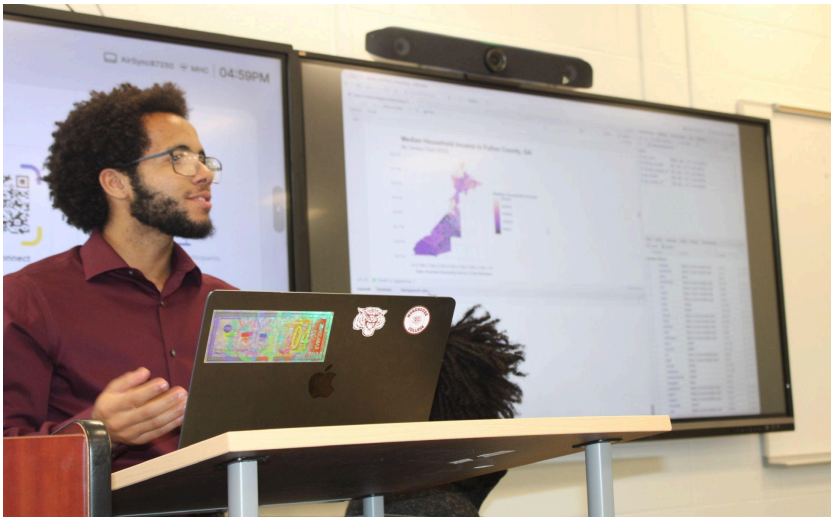


Figure 5: Elijah Qualls (TA) leading a portion of the student workshop, explaining how students are expected to generate the same type of map which is pictured on the screen.

Once the demonstration was over, the attendees were assigned different variables to map in Fulton county. These maps are intended to be used within HJ Lab's larger research project pertaining to flock cameras in Atlanta, GA. All of the variables were selected from open-source census data. Some of the assigned variables included internet access, vehicle availability, commute times, race, and policing tendencies. This task presented the opportunity for students to take the code which was introduced to the group and navigate it individually. This served as an ultimate test of comprehension and retention. A student navigating the data set, finding the specific variable code, calling that code, making new objects with the census and spatial data, and then graphing said objects onto a map of Fulton County indicates major growth, considering most attendees had little to no experience with R prior to this workshop. The below figure conveys the average reported change in levels of student mastery before and after the demonstration, which then facilitated the creation and submission of their own maps using unique census variables.

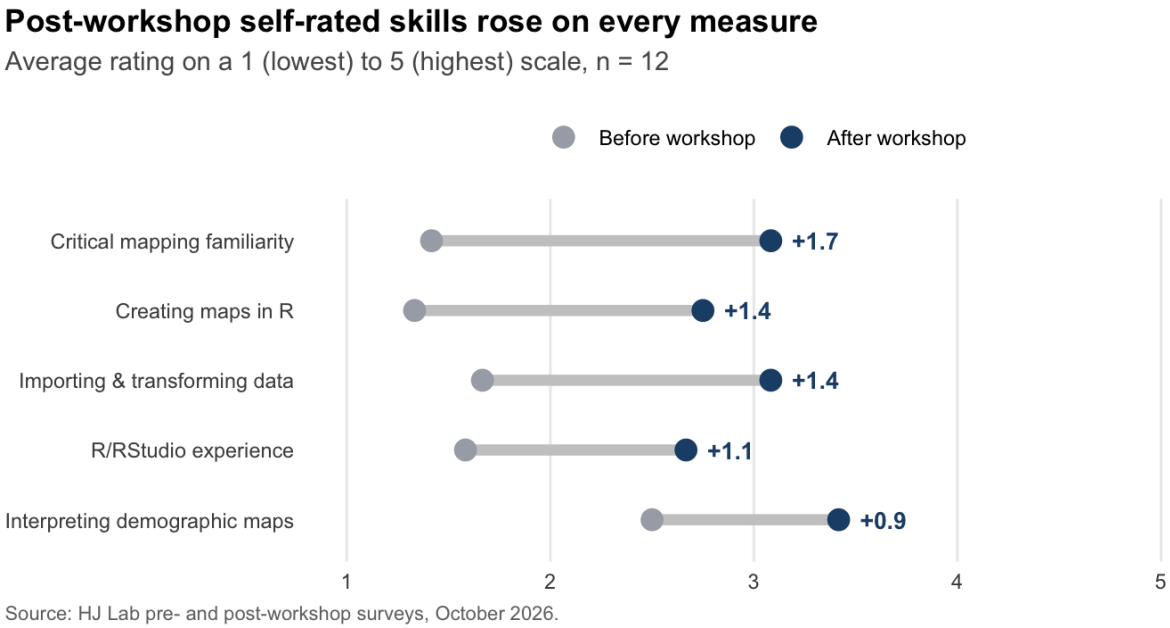


Figure 6

III. Outcome

The proliferation of the computational geographical skills of participants were not only measured through the completion of the assignment. Pre and post surveys were administered to the students in efforts to measure perceived growth.

Figure 7 shows the responses from 10 out of the 14 attendees. This question specifically asks about their familiarity with Critical Census Geography (CCG). On average, there is a significant increase in participants' familiarity from before the workshop to after. This figure illustrates the session's success in introducing more students to R critical mapping methodology using open-source census data.

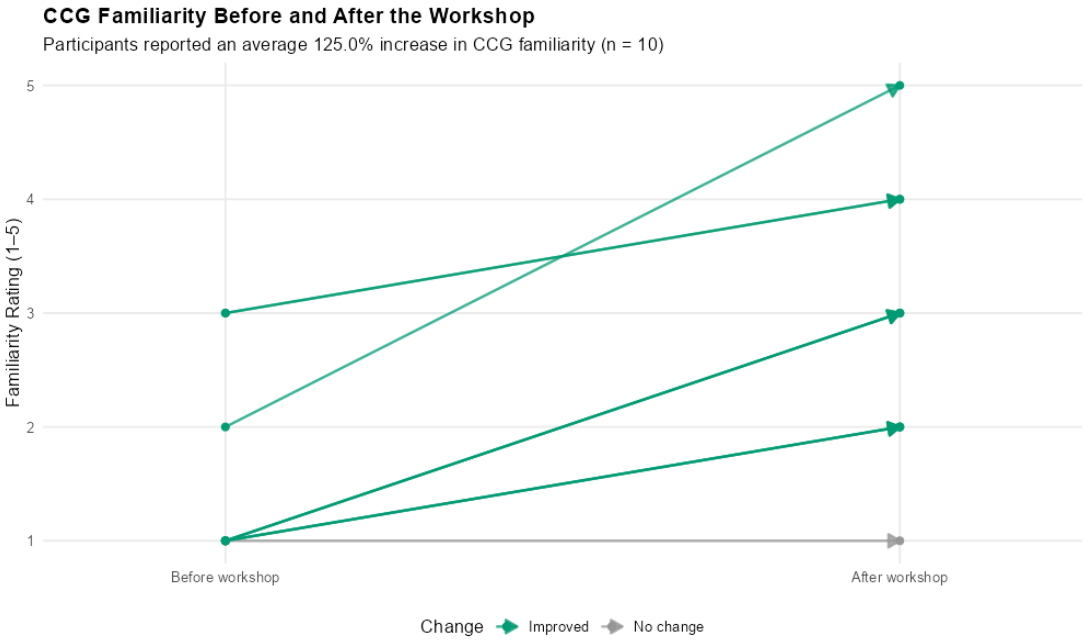


Figure 7

Furthermore, *Figure 8* shows that for any given category, 83% or more students agreed with the sentiment portrayed in the post-survey questions. The lab proved to be effective in teaching students how to use R, encouraging them to recommend other students to attend future workshops hosted by HJ Lab, and pursue critical mapping projects in the future.

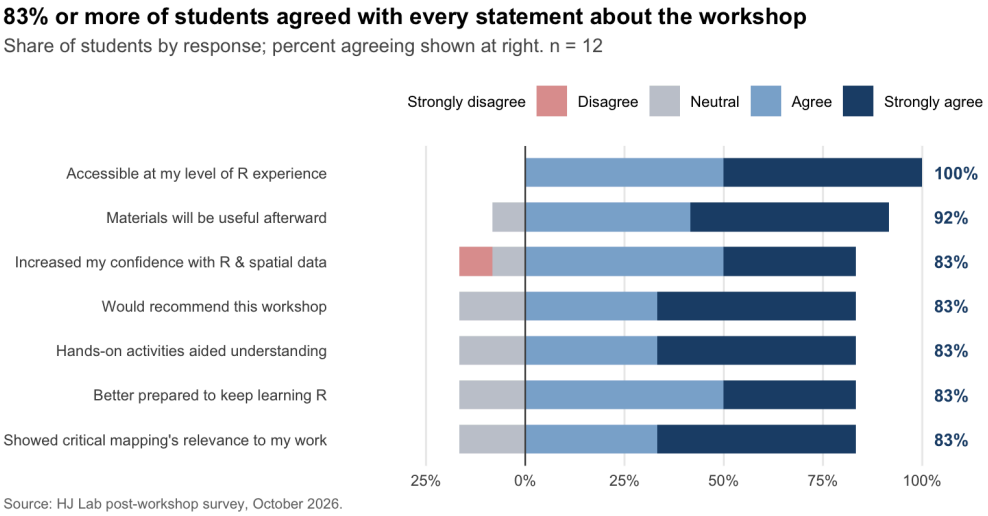


Figure 8

Figure 9 depicts students developing confidence in using their newfound skills in a variety of research environments such as class assignments, and conference presentations, further illustrating the success and lasting impact of the student workshop.

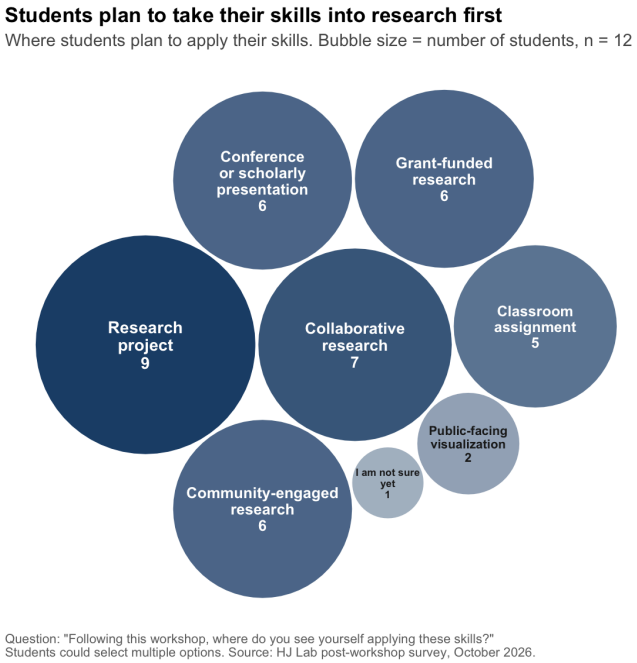


Figure 9

IV. Conclusion

The Open Critical Mapping Workshop provided substantial benefits for all parties involved that will certainly impact their future academic and professional endeavors. First, faculty from different departments learned a new method to apply to their areas of research, further pursuing the interdisciplinary nature of Morehouse College. Second, the students in attendance were introduced to a new coding software that most of them were unfamiliar with, including the computer science majors. This offers opportunities for the students to explore the possibilities of data science and engage with the intersection of political science and computer science within personal projects. Lastly, the research assistants gained hands-on experience in providing support to new R users which indirectly improved the mastery level of the research/teaching assistants.

Looking ahead, students and faculty alike have communicated aspirations to start new projects using open-source census and spatial data to map various developing paradigms. For example, the HJ Lab is currently working on running geographic models and overlaying those maps with the open-source flock camera tracker maps. We are working to find spatial correlations between where the cameras are located (indicating heavy surveillance) and various potential variables. Some of these variables are in the early stages of being analyzed, thanks to the help of some of the attendees of the student workshop. Furthermore, faculty members have also voiced interest in mapping different variables which directly relate to their areas of research. For example, Dr. Keith Hollingsworth wants to map Black businesses in Atlanta over time, possibly dating his visualizations back to the Atlanta Race Massacre in 1906. Dr. Tina

Chang also said that she is interested in mapping public spaces such as parks and public sitting areas across Atlanta to track the trends.

A future consideration for the lab driven by participant survey data is to host more workshop series using R, building on the programming and geospatial analysis skills learned in the first demonstration. As seen in *Figure 10*, there is 100% demand for more guided learning experiences in a supportive environment where students can build confidence practicing open research methods using R. A prospective next workshop could expound on the applications of computational geography and spatial data, potentially introducing common geospatial machine learning algorithms such as *K*-Nearest Neighbors and *K*-Means Clustering.

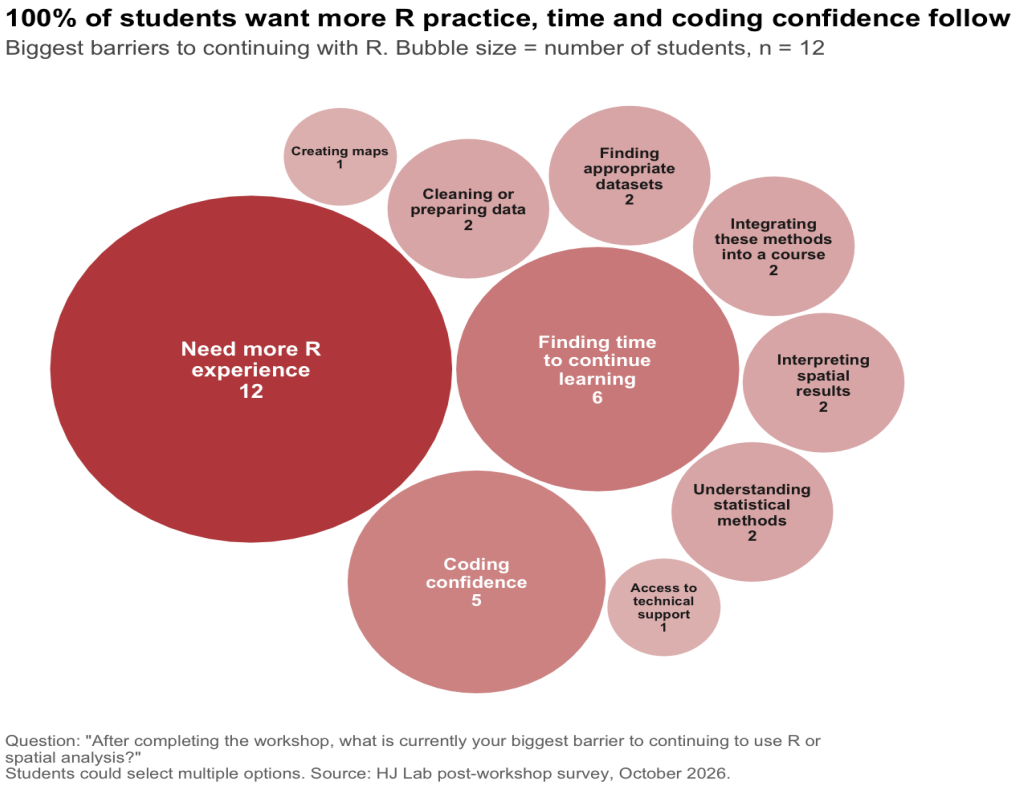


Figure 10

Without the ORCA CAP Award, students and faculty would not have grown within their understanding of open-source material. This workshop has not only sparked inspiration, but it has also sparked a passion within students to code not simply for an assignment's sake, but for personal pleasure and understanding.